THE PROBLEM-SOLVING TOOLBOX!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use the tool the question names. Check your answer.

No.S1

Use a strip. Kumar has 12 marbles. Ravi has 5 fewer marbles than Kumar.



(a) Ravi has ☐ marbles.

(b) They have ☐ marbles in all.

No.S2

Do one step at a time. A shop has 30 apples. It sells 12 apples in the morning and 9 apples in the afternoon.

(a) After the first step, ☐ apples are left.

(b) After the second step, ☐ apples are left.

No.S3

Make a table. Each row has 3 more beads than the row before. Fill in the table.

Row	1	2	3	4	5	6
Beads	3	6	9	☐	☐	☐

(a) Row 5 has ☐ beads.

(b) Row 6 has ☐ beads.

Name _____ Date _____

Use the tool the question names. Check your answer.

No.S4

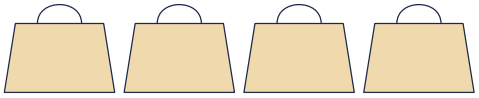
Work backwards. Ravi thinks of a number. He takes away 8 and gets 17.

(a) Ravi's number is ☐ .

(b) Check: Ravi's number take away 8 is ☐ .

No.S5

Find one first. 4 bags hold 20 apples. All the bags hold the same number of apples.



(a) 1 bag holds ☐ apples.

(b) 7 bags hold ☐ apples.

No.S6

Use a list. Ali has three coins: 5¢, 10¢ and 20¢. He picks 2 coins. Fill in the blanks.



PLAY MONEY

(a) 5¢ and 10¢ make ☐ ¢.

(b) 5¢ and 20¢ make ☐ ¢.

(c) 10¢ and 20¢ make ☐ ¢.

(d) There are ☐ different pairs of coins.

Name _____ Date _____

The tool is not named: you choose. Choose any question, in any order.

No.T1

Read the story. 3 rulers cost $15. All the rulers cost the same. Siti buys 5 rulers.

(a) Circle the tool that fits best. **Table** **List** **Find one first**

(b) 5 rulers cost $ ☐ .

No.T2

Ben pays with a $10 note for a toy and a card. The toy costs $6. His change is $1.

(a) Ben spends $ ☐ in all.

(b) The card costs $ ☐ .

No.T3

Siti is 4th from the front of a line. She is also 3rd from the back. How many children are in the line? Use the box to show your thinking.

(a) ☐ children are in front of Siti.

(b) There are ☐ children in the line.

No.T4

Mei has 7 coins. Some are 5¢ coins and some are 10¢ coins. The coins make 50¢ in all. You may use the table.

10¢ coins	5¢ coins	Total

(a) Mei has ☐ coins of 5¢.

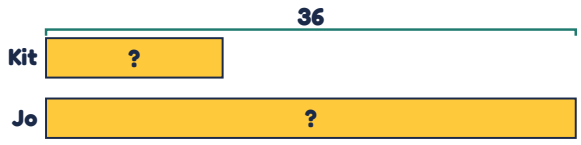
(b) Mei has ☐ coins of 10¢.

Name _____ Date _____

Check your answer a second way. Choose any question. You do not have to finish them all.

No.G1

Jo has 3 times as many stickers as Kit. They have 36 stickers in all. Solve it two ways: a strip and a table of tries.



Kit	Jo	In all
8	24	32
10	30	40
9	☐	36

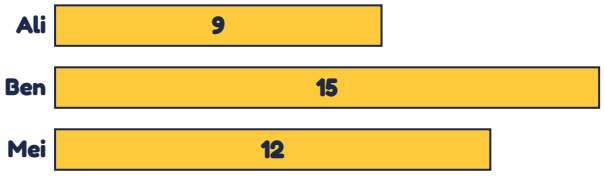
(a) Road 1, the strip: Kit has ☐ stickers.

(b) Road 2, the table: when Kit has 9, Jo has ☐ stickers.

(c) Do both roads give the same answer? Circle Yes or No. **Yes** **No**

No.G2

Ali has 9 stickers, Ben has 15 stickers and Mei has 12 stickers. Only Ben and Ali move stickers. Then all three children have the same number.



(a) Each child ends with ☐ stickers.

(b) Ben gives ☐ stickers to Ali.

No.G3

Siti reads 4 pages on day 1, 7 pages on day 2 and 10 pages on day 3. Each day she reads 3 more pages than the day before. Finish the sentence about the tool you used.

(a) She reads 22 pages on day ☐ .

(b) I used _____ because _____ . (Many answers are right.)

Name _____ Date _____

Investigation: Pick a tool

First: Pick one tool card. Write its name: _____

Next: Write a short problem that fits the tool. Solve it. Then check it a second way. Write your problem and your answer.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Mei thinks of a number. She adds 4 and gets 15.

Circle the tool that fits best. **Table** **Work backwards** **List**

No.E2

Use a strip. Ali has 11 stickers. Mei has 4 fewer stickers than Ali.

(a) Mei has ☐ stickers.

(b) They have ☐ stickers in all.

No.E3

5 buns cost $20. A child in another class says 1 bun costs $5.

Circle True or False. **True** **False**

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ picking a tool ☐ using a strip ☐ working backwards ☐ something else

What helped me: ☐ the tool cards ☐ drawing it ☐ checking a second way ☐ I was not stuck

Next time I will try: ☐ picking a tool ☐ checking a second way ☐ saying why ☐ asking a question

☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

Kit page: tool and problem cards

A grown-up cuts out the 15 cards before the lesson: 7 tool cards and 8 problem cards. Each problem card has a short label; your tutor reads the whole problem aloud. No cutting? Write the 7 tool names on 7 scraps of paper.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

Tool cards and problem cards: a grown-up cuts out all 15.



Strip



Table



List



Draw it



Work backwards



Find one first



Steps

1. Ali 9, Mei 4 fewer

2. Rows of 2, 4, 6 blocks

3. Pairs from 3 fruits

4. Ravi 3rd in a line of 7

5. A number take away 6 is 9

6. 5 buns cost $15

7. 20 sweets, 8 eaten, 5 bought

8. 2 red beads, 3 blue